

AS Level Mathematics — Pure Mathematics 1 (P1)

Practice Paper 5

Time allowed: **1 hour 50 minutes**

Instructions to candidates

- Answer **all** questions. Write your answers on paper, showing all necessary working. Marks may be lost for insufficient working.
- If a numerical answer is not exact, give it correct to **3 significant figures**; give angles in degrees correct to **1 decimal place**, unless a different level of accuracy is specified in the question.
- A scientific calculator may be used. Do **not** use a graphical or symbolic calculator.
- The number of marks is given in brackets [] at the end of each question or part question. The total mark for this paper is **75**.
- This paper is written in the style of Cambridge International 9709 Paper 1 (Pure Mathematics 1). It is a practice paper, not an official examination paper.

List of formulae (as provided in the examination)

Quadratic equation: for $ax^2 + bx + c = 0$, $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Binomial expansion: $(a + b)^n = a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \dots + b^n$, where $\binom{n}{r} = \frac{n!}{r!(n-r)!}$

Arithmetic series: $u_n = a + (n-1)d$, $S_n = \frac{1}{2}n(a + l) = \frac{1}{2}n\{2a + (n-1)d\}$

Geometric series: $u_n = ar^{n-1}$, $S_n = \frac{a(1-r^n)}{1-r}$ ($r \neq 1$), $S_\infty = \frac{a}{1-r}$ ($|r| < 1$)

Circular measure: arc length = $r\theta$, sector area = $\frac{1}{2}r^2\theta$ (θ in radians)

Trigonometry: $\tan \theta \equiv \frac{\sin \theta}{\cos \theta}$, $\sin^2 \theta + \cos^2 \theta \equiv 1$

Differentiation and integration: $\frac{d}{dx}(x^n) = nx^{n-1}$, $\int x^n dx = \frac{x^{n+1}}{n+1} + c$ ($n \neq -1$)

1 Express $x^2 + 5x - 2$ in the form $(x + a)^2 + b$, where a and b are constants, and hence state the minimum value of $x^2 + 5x - 2$. **[3]**

2 Solve the equation $2 \sin^2 \theta + 3 \cos \theta = 0$ for $0^\circ \leq \theta \leq 360^\circ$. **[4]**

3

(a) Find the first four terms, in ascending powers of x , in the expansion of $(1 + 2x)^5$. **[2]**

(b) Hence find the coefficient of x^3 in the expansion of $(1 - 3x)(1 + 2x)^5$. **[2]**

4 A geometric progression has first term a and common ratio r , where $r > 0$. The sum to infinity of the progression is three times the first term.

(a) Find the value of r . [2]

(b) The sum of the first two terms is 20. Find the sum of the first six terms, giving your answer correct to 3 significant figures. [3]

5 A sector OAB of a circle with centre O and radius r cm has angle θ radians. The area of the sector is 24 cm^2 and the length of the arc AB is 8 cm.

(a) Find the values of r and θ . [3]

(b) Find the area of the segment bounded by the arc AB and the chord AB . [3]

6 The line $3x + 4y = 24$ meets the x -axis at the point P and the y -axis at the point Q .

(a) Find the length of PQ . [2]

(b) Find the equation of the circle which has PQ as a diameter, and determine whether the origin O lies on this circle. [4]

7 Functions f and g are defined by

$$f(x) = 5 - 2x \quad \text{for } x \in \mathbb{R}, \quad g(x) = x^2 + 1 \quad \text{for } x \geq 0.$$

(a) Find an expression for $f^{-1}(x)$. [2]

(b) Find an expression for $fg(x)$ and state the range of fg . [2]

(c) Solve the equation $f^{-1}(x) = g(x)$. [3]

8 A curve has equation $y = x^2 + \frac{16}{x}$ for $x > 0$.

(a) Find the coordinates of the stationary point of the curve and determine its nature. [4]

(b) A point moves along the curve in such a way that its x -coordinate is increasing at a constant rate of 0.05 units per second. Find the rate of change of the y -coordinate at the instant when $x = 4$. [3]

9

(a) Prove the identity $\cos^4 \theta - \sin^4 \theta \equiv 1 - 2 \sin^2 \theta$. [3]

(b) Hence solve the equation $\cos^4 \theta - \sin^4 \theta = \frac{1}{2}$ for $0^\circ \leq \theta \leq 180^\circ$. [3]

(c) Solve the equation $\tan(2x - 30^\circ) = 1$ for $0^\circ \leq x \leq 180^\circ$. [2]

10 The curve $y = x^2$ and the line $y = x + 2$ meet at the points A and B .

(a) Find the coordinates of A and B . [2]

(b) Find the area of the region enclosed by the curve and the line. [3]

(c) The region bounded by the curve $y = x^2$, the y -axis and the line $y = 4$ (for $x \geq 0$) is rotated through 360° about the y -axis. Find the volume of the solid formed, giving your answer in terms of π . [3]

11

(a) The third term of an arithmetic progression is 11 and the sum of the first 12 terms is 300.

(i) Find the first term and the common difference. [3]

(ii) Find the number of terms of the progression that are less than 200. [2]

(b) The first three terms of a geometric progression are k , $k + 6$ and $4k$, where k is a non-zero constant. Find the possible values of k and, for each value, the corresponding common ratio. [4]

12 A circle with centre C has equation $(x - 3)^2 + (y + 1)^2 = 20$. The line $y = 2x + 1$ meets the circle at the points A and B .

(a) Find the coordinates of A and B . [3]

(b) Find the coordinates of M , the midpoint of AB . [2]

(c) Show that CM is perpendicular to AB . [3]

END OF PAPER (Total: 75 marks)